Configuring VPC Peering Between Two VPCs

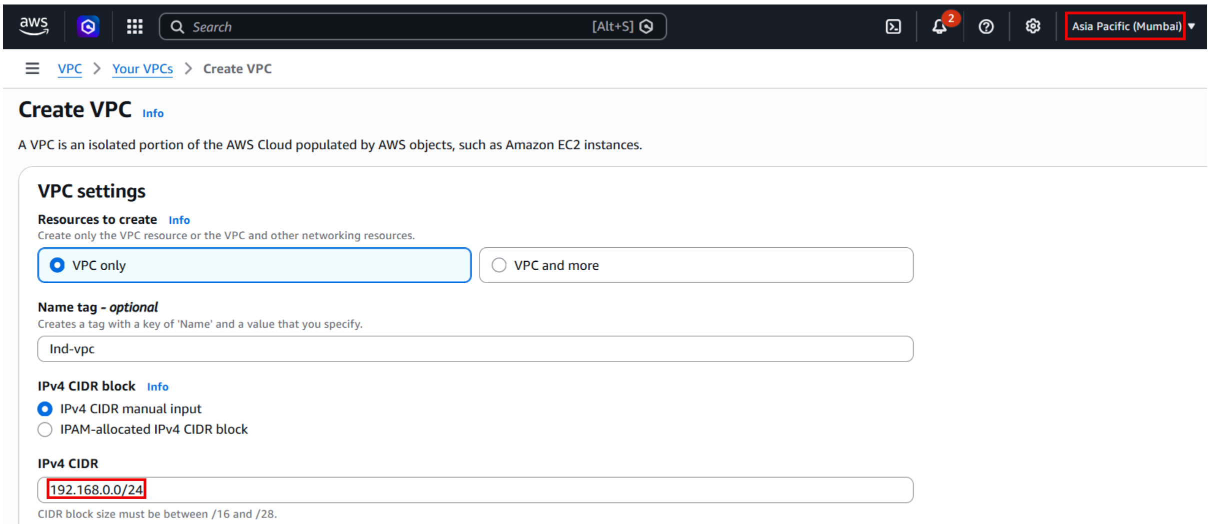
To Begin with the Lab

Summary of the Lab

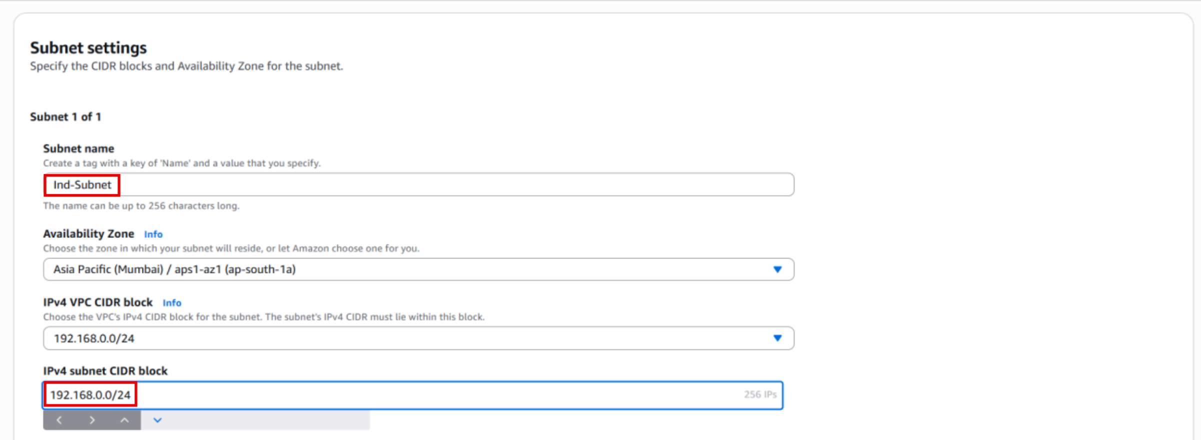
This lab demonstrated how to configure VPC Peering between two VPCs located in different AWS Regions (Mumbai and US East–Ohio) to enable private communication. Separate VPCs with non-overlapping CIDR blocks, subnets, and EC2 instances were created without public IPs. Initial connectivity between the instances failed because the VPCs were isolated. A VPC Peering Connection was then created, accepted, and activated, followed by updating the route tables in both VPCs to route traffic through the peering connection. After the configuration, the EC2 instances successfully communicated using their private IP addresses over the AWS network without requiring an Internet Gateway or public internet access.

Note: This lab demonstrates communication between two VPCs located in different AWS Regions. Both VPCs are created in the same AWS account, but the same process also works for VPCs in different AWS accounts.

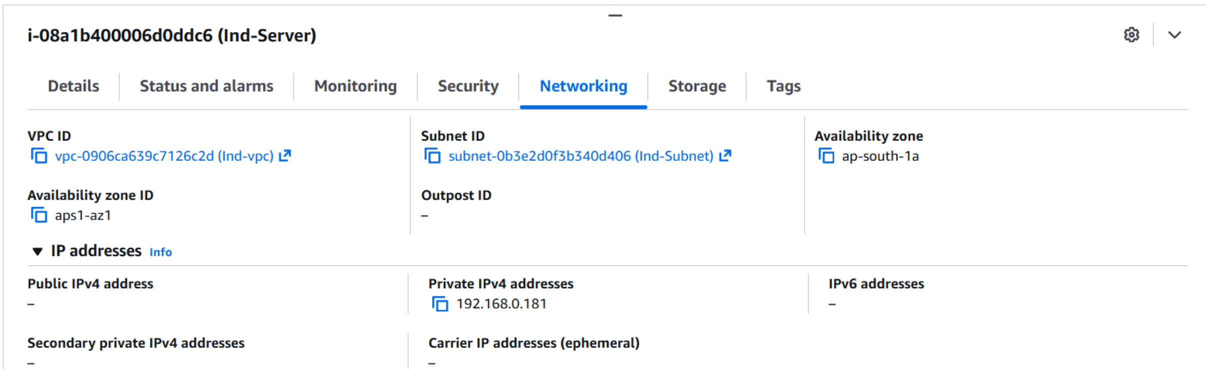
Open the **AWS Management Console** and switch to the **Mumbai (ap-south-1)** region. Create a new VPC named **Ind-VPC** with the IPv4 CIDR block **192.168.0.0/24**, ensuring that its address range is unique.



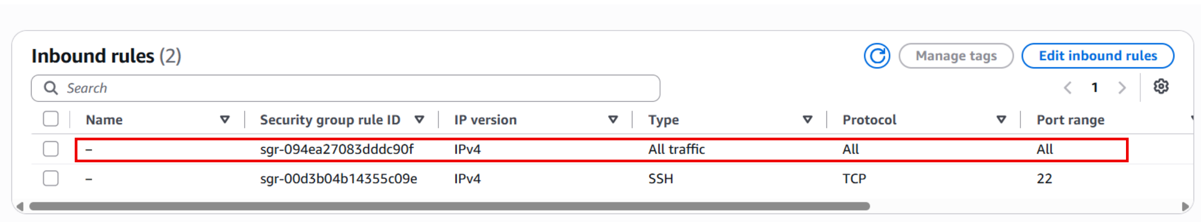
Create a subnet inside **India-VPC** by selecting the VPC, assigning a name such as **India-Subnet**, choosing the **ap-south-1a** Availability Zone, and using the entire CIDR block **192.168.0.0/24**. Since no Internet Gateway is configured, this subnet functions as a private subnet.



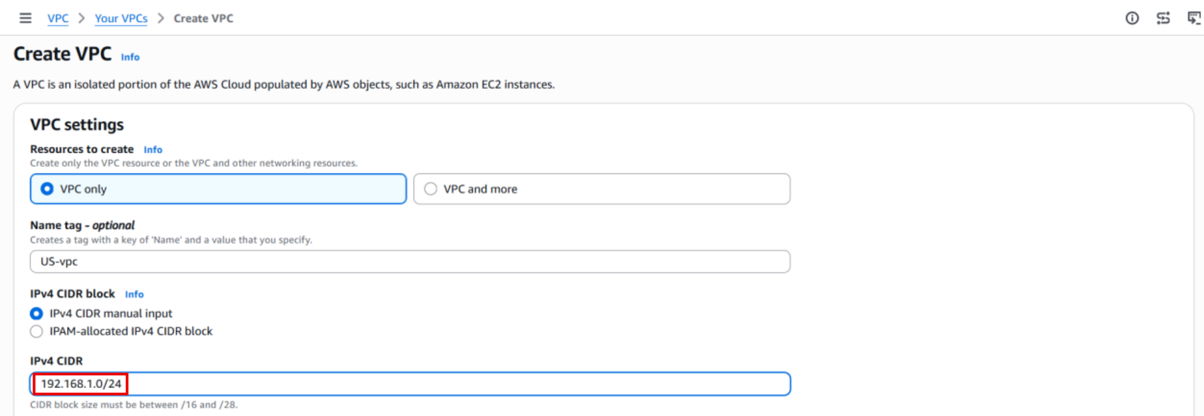
Launch an EC2 instance named **India-Server** inside the India subnet. Disable **Auto-assign Public IP** and configure the Security Group to allow **All Traffic** so that connectivity testing between the VPCs is not blocked by firewall rules.



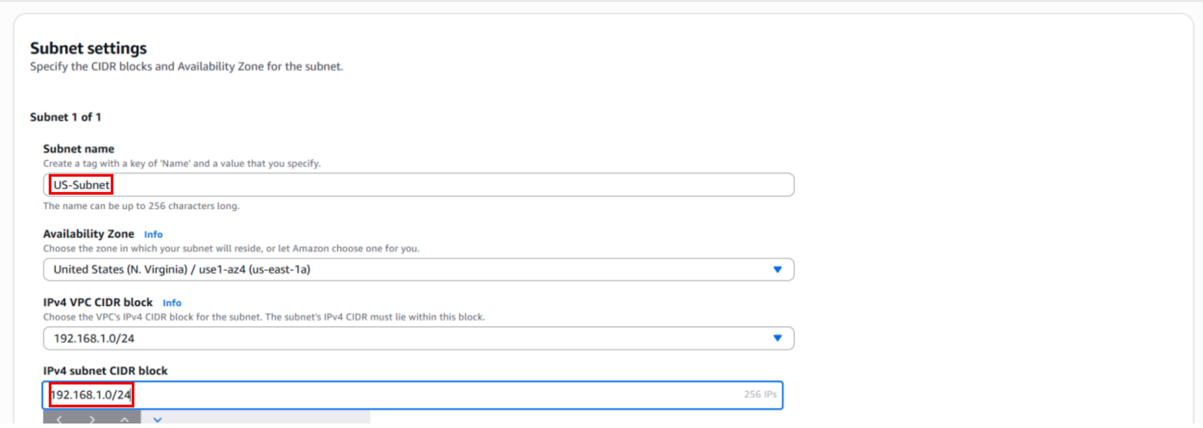
Security Group inbound rule for US-server



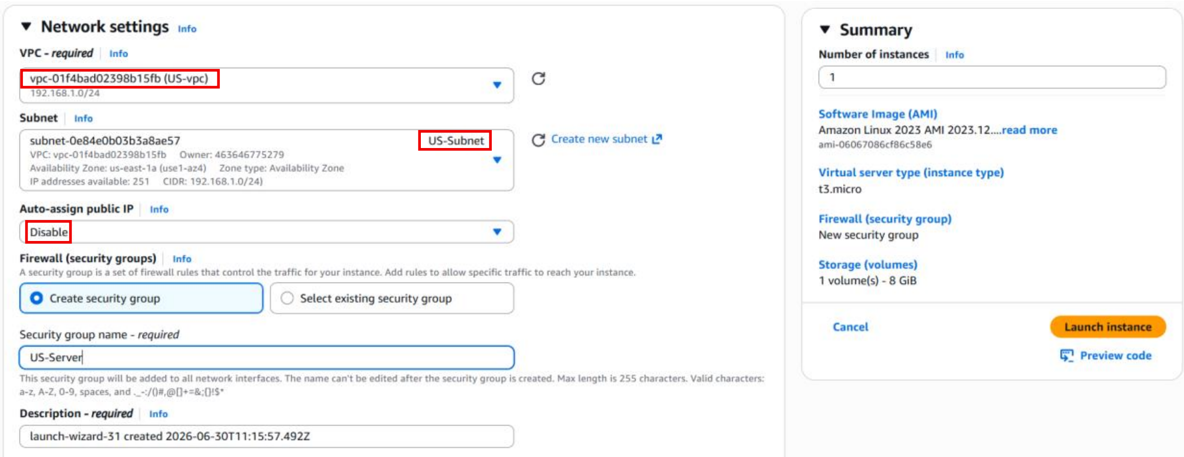
Switch to the **US East (Ohio)** region and create another VPC named **US-VPC** using a different CIDR block such as **192.168.1.0/24**. The CIDR ranges of both VPCs must be different because overlapping CIDR blocks are not supported for VPC Peering.



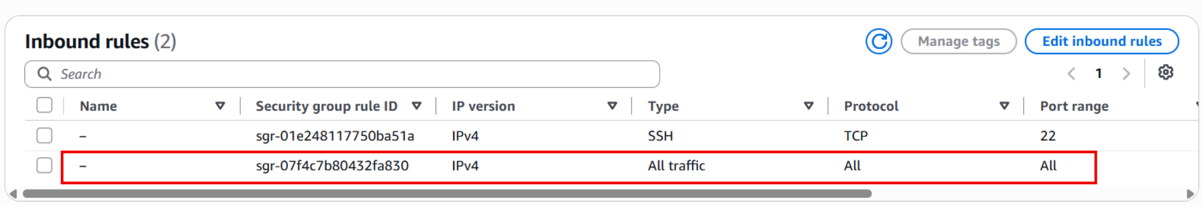
Create a subnet inside **US-VPC**, assign it a name such as **US-Subnet**, choose the appropriate Availability Zone, and use the VPC CIDR block **192.168.1.0/24**.



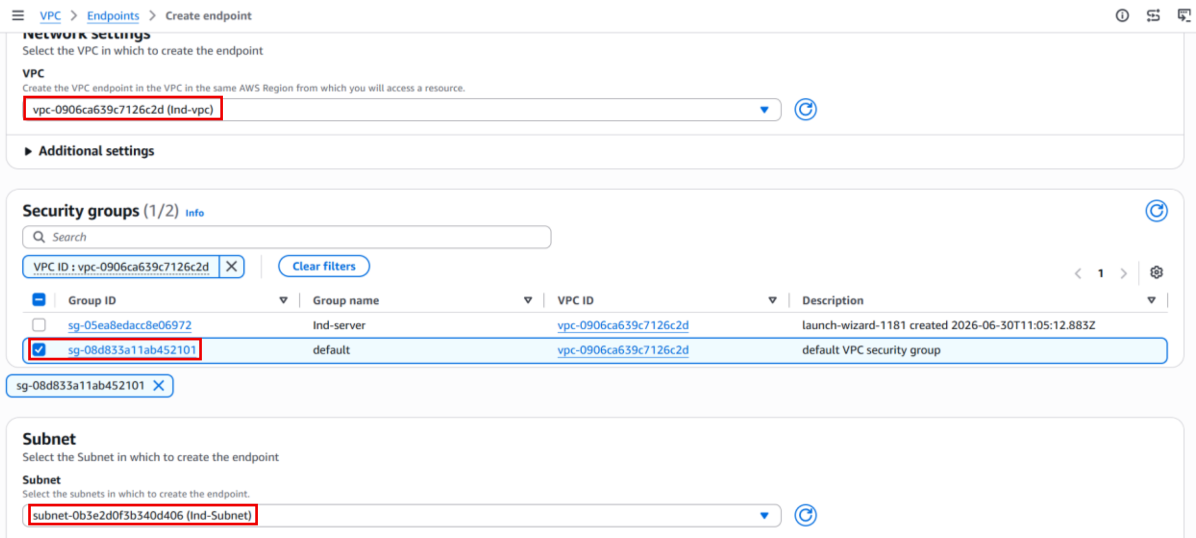
Launch another EC2 instance named **US-Server** inside the US subnet. Disable **Auto-assign Public IP** and configure the Security Group to allow **All Traffic**, just as you did for the India server.



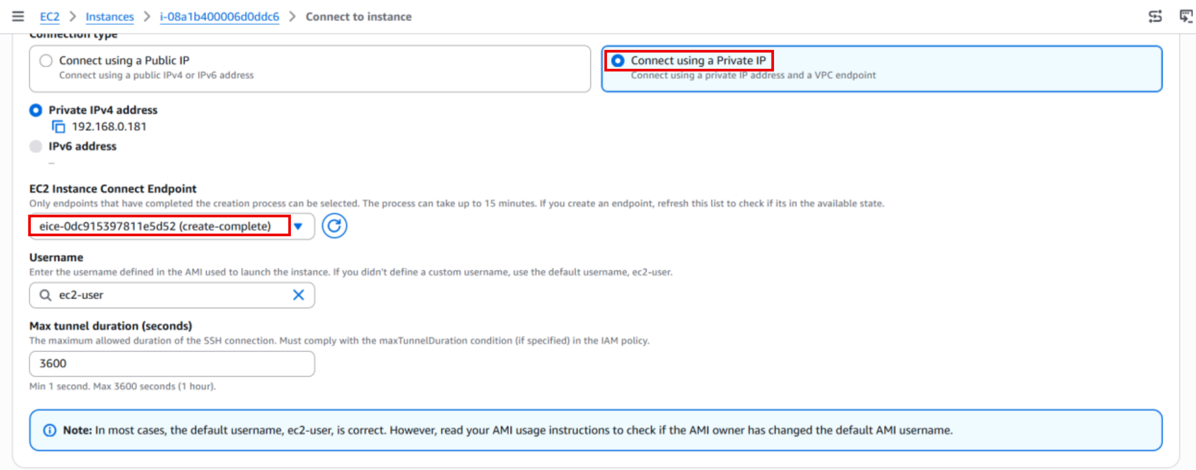
Security Group inbound rule for US-server



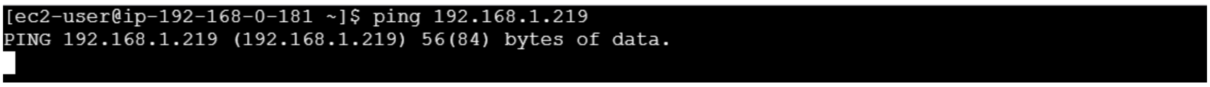
Since both EC2 instances are located in private subnets, create an **EC2 Instance Connect Endpoint** for the India VPC. After the endpoint becomes **Available**, use it to connect to the India EC2 instance through the AWS Console.



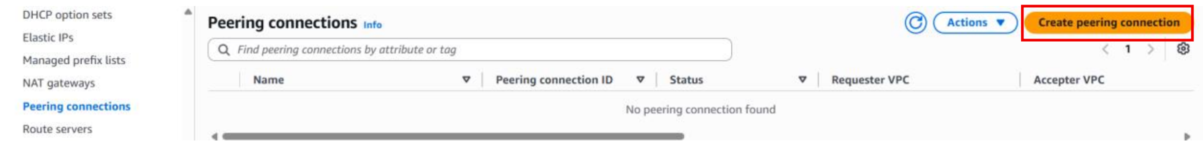
From the India EC2 instance, attempt to ping the **Private IP Address** of the US EC2 instance. The ping request will fail because, by default, separate VPCs cannot communicate with each other, even if they belong to the same AWS account.

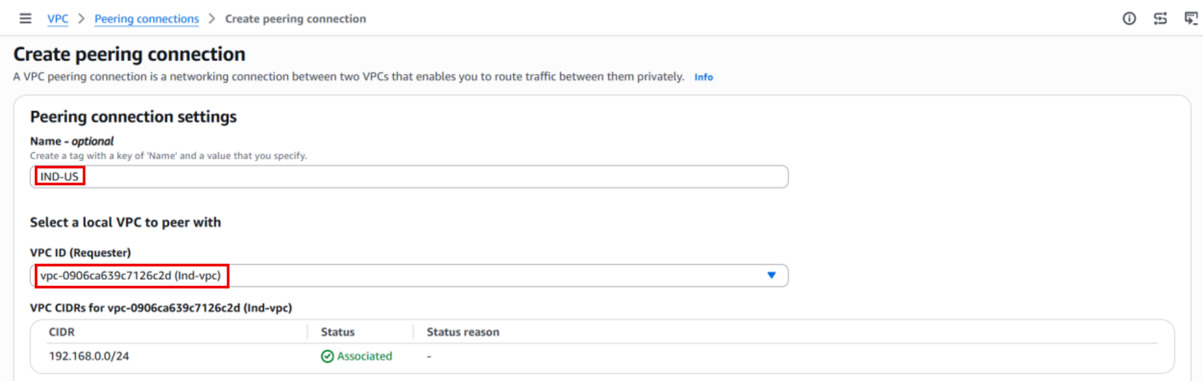


Now after the connecting to the instance, we will try to ping the private IP address of US-Server



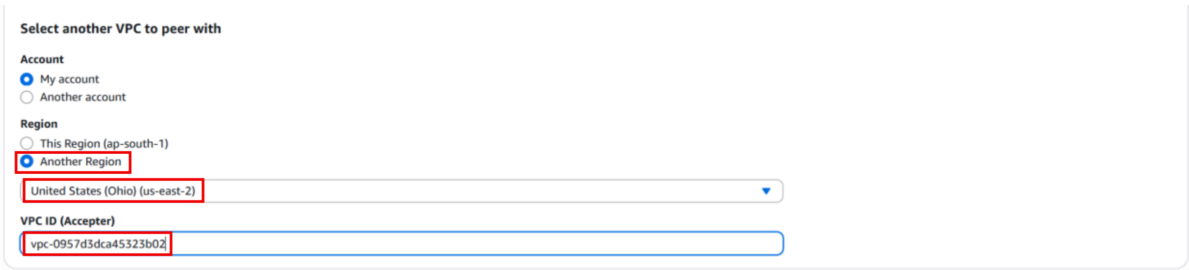
Return to the **Mumbai** region and open the **Peering Connections** section in the VPC console. Click **Create Peering Connection**, provide a name such as **Ind-US**, select **India-VPC** as the requester VPC,





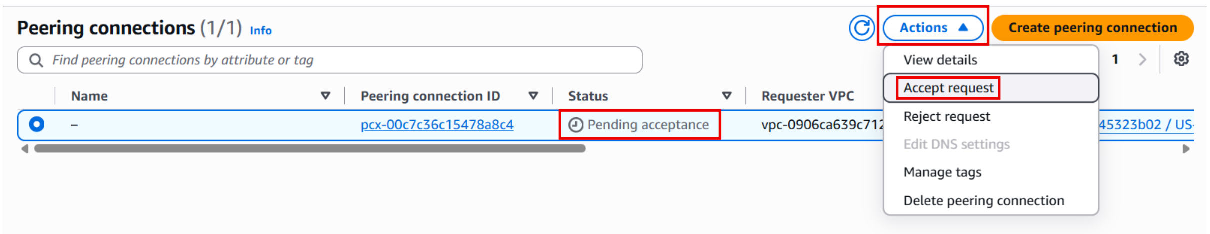
choose the **US East (Orio)** region, and select **US-VPC** as the accepter VPC.

Copy the VPC Id from the US region where we have created vpc.

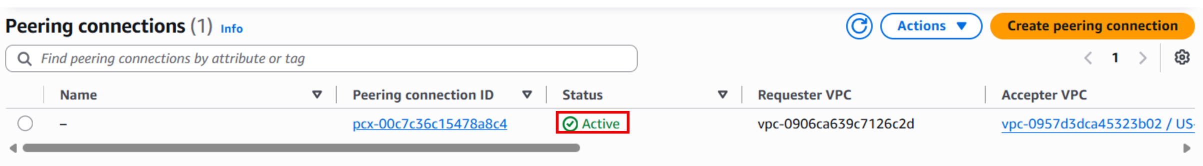


Click Create Peering Connection

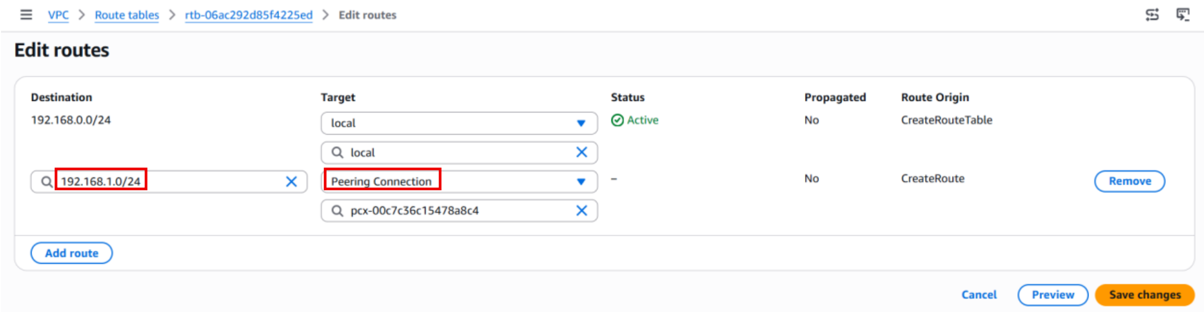
Switch to the **US East (Ohio)** region, open **Peering Connections**, select the pending request, and click **Accept Request**.



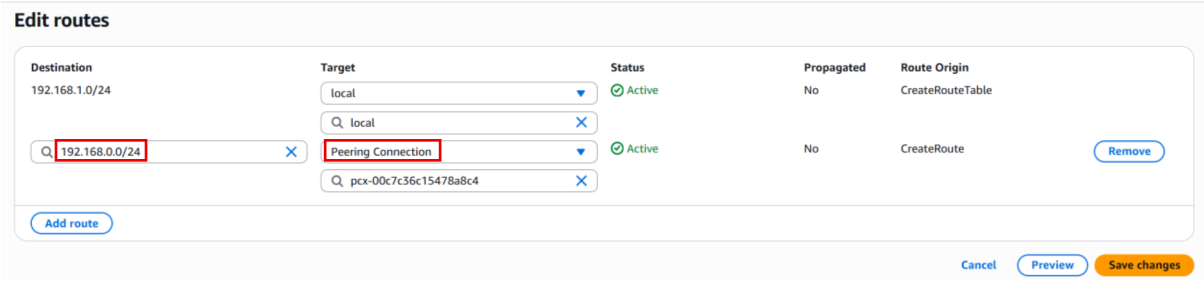
Wait until the peering connection status changes to **Active** in both regions.



Open the **Route Table** associated with **India-VPC** and edit its routes. Add a new route with the destination **192.168.1.0/24** and select the newly created **VPC Peering Connection** as the target, then save the changes.



Switch to the **US East (Ohio)** region and edit the **US-VPC Route Table**. Add another route with the destination **192.168.0.0/24** and choose the same **VPC Peering Connection** as the target, then save the route.



Verify that communication is occurring entirely through **Private IP Addresses** over the AWS network. No Internet Gateway or Public IP Address is required for communication between the two VPCs after peering has been configured.